Simulated Targeted Beacon from 61 Cygni - Link & Detection Report

Scenario: a tidally-locked habitable-zone planet at 0.39 AU around 61 Cyg A transmits a 1-Hz wide beacon at the 21-cm hydrogen line (1420.405751 MHz) with EIRP = 3e10 W - near the Breakthrough Listen GBT detection floor for this distance. Receiver: GBT-class (SEFD 10 Jy), 300 s integration.

Quantity	Value
Distance	3.496 pc / 11.4 ly
Frequency	1420.405751 MHz (1 Hz wide)
Transmitter EIRP	3e10 W (Kardashev K = 0.50)
Received flux density	20.5 Jy in 1 Hz
GBT detection floor (10 sigma, 300 s)	~4.1 Jy
Post-integration SNR	50 (detection = rock-solid)
Orbital velocity of planet	39.3 km/s
Max orbital Doppler drift	-0.125 Hz/s
Frequency sweep in one 300 s obs	-37.5 Hz (across 37 fine channels)
Coherent-sim mean SNR	140 (target 50: conservative model margin)

Interpretation

- A civilization even slightly ahead of ours (K~0.5) beaming deliberately toward us is trivially detectable - the received 20.5 Jy is five times the GBT 10-sigma floor.

- The Doppler signature is the smoking gun: a steady -0.125 Hz/s drift from orbital acceleration is exactly what SETI pipelines (turboSETI-style de-drift search) flag as non-terrestrial, since RFI sits still in the topocentric frame.

- Because the planet is tidally locked, rotation adds almost nothing - the drift tracks the 108-day orbit, repeating identically each period: a repeating, drifting, narrowband signal is the classic 'wow that's interesting' profile.

- Files: beacon_61cyg_timeseries.csv (300 one-second cells: peak frequency + excess power), beacon_61cyg_sonified.wav (audible clone: 880 Hz carrier, drift scaled x60, 1s-on/1s-off beacon gating over faint noise).